

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer Examination – 2024

Course: TY B. Tech.

Branch: Mechanical Engineering

Semester: VI

Subject Code & Name: Manufacturing Processes-II (BTMC 601)

Max Marks: 60

Date: 13 / 06 / 2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Describe with neat sketch external cylindrical grinder.	CO4	6
B)	Discuss Grit, Grade, and Structure of wheel.	CO4	6
C)	Write a note on selection of grinding wheel.	CO5	6
Q.2	Solve Any Two of the following.		12
A)	Explain with neat sketch geometry of single point cutting tool.	CO2	6
B)	Discuss with neat sketch various chips formed during metal cutting.	CO2	6
C)	At a cutting speed of 30 mpm, a cutting tool had a life of 55 minutes between regrinds when doing rough cutting of mild steel. Calculate the tool life when engaged on light finishing cuts. Assume $n=1/8$ and $1/10$ for rough and finishing cuts respectively.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Discuss temperature distribution in metal cutting.	CO3	6
B)	What are the factors affecting the tool life.	CO3	6
C)	List out different types of cutting tool materials, explain any two.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	What is power metallurgy? Explain applications of powder metallurgy components.	CO4	6
B)	Describe with neat sketch Isostatic pressing in powder metallurgy.	CO1	6
C)	Explain presintering and sintering in powder metallurgy.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Explain with neat sketch extrusion process in processing of plastic.	CO2	6
B)	Describe with neat sketch blow molding process.	CO4	6
C)	Explain with neat sketch Injection molding process.	CO4	6
